SurakṣāMārga.ai (SuraksaMarga.ai)

5G INNOVATION HACKATHON 2026

A 5G-ENABLED AI SYSTEM FOR WOMEN SAFETY NAVIGATION

SAFETY OVER SPEED.

Great navigation should feel protective.

We re-engineer urban routing around the realities women face, weighing 157,160 real crime records to navigate around danger, not just toward a destination.

01
Safe Route Engine

02
Crime-Weighted Risk

03
5G SOS Layer

04
Live Safety Loop

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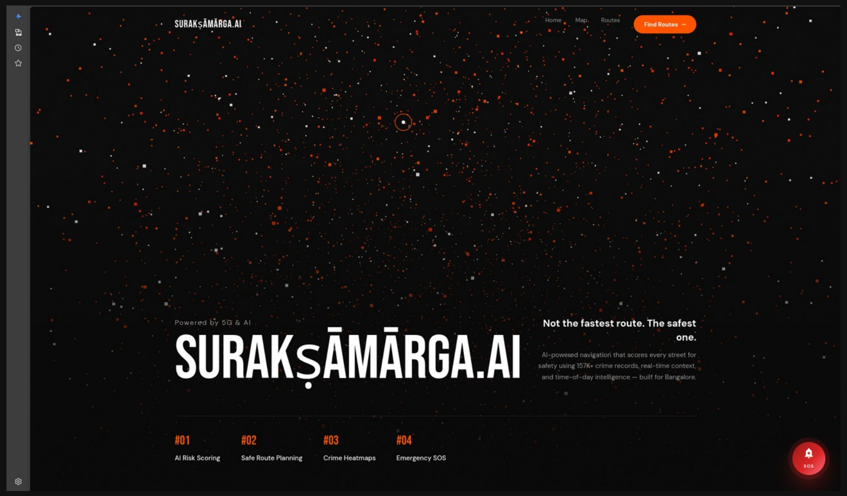
01 INTRODUCTION

NAVIGATION WAS NEVER BUILT FOR HER.

Modern mapping apps optimise for journey time, exposing women to isolated shortcuts, desolate backstreets and unlit alleys, especially after dark.

SuraksaMarga.ai bridges geospatial routing with real-world danger prevention. It actively protects women by prioritising localised safety intelligence over arbitrary travel speed.



02 PROBLEM STATEMENT

THE MAP IGNORES WHAT WOMEN ALREADY KNOW.

01

Time-Only Models

Commercial systems apply rigid pedestrian models focused entirely on reducing time and distance, never on safety.

02

Forced Unsafe Shortcuts

Women are routed into inherently threatening terrain without any advanced warning of crime hotspots or isolation.

03

No Protective Layer

Users must exit the map to formulate emergency communications when danger is imminent, a critical systemic failure.

03 OBJECTIVES

RAPID. SECURE. WOMEN-FIRST TRANSIT.

01

Safe Route Recommendation

Map the safest traversal, explicitly avoiding identified danger zones.

02

Women-Centric Crime Weighting

Penalise routes featuring histories of gender-based crime incidents.

03

SOS Emergency System

Trigger live location tracking and responder alerts seamlessly.

04

5G Low-Latency Capability

Utilise 5G for immediate safety-system distress broadcasts.

04 EXISTING WORK & RESEARCH GAP

FAST, NOT SAFE. WE BRIDGE THE GAP.

Google Maps and OSRM rely exclusively on the "fastest route" mentality. They possess no capacity to distinguish a secure, well-lit main avenue from an inherently unsafe, desolate alleyway.

Gender-specific risk is totally absent from modern mapping tools. Existing solutions treat all pedestrians uniformly. SuraksaMarga.ai directly enforces safety constraints first and speed constraints second.

	EXISTING	OURS
ROUTING BASIS	Time / distance	Crime-weighted safety
GENDER AWARENESS	None	Women-first weighting
NIGHT-TIME LOGIC	Static	Active multipliers
SOS INTEGRATION	External app	Built into route
LATENCY TARGET	Best-effort	5G edge ~10 ms

05 PROPOSED METHODOLOGY

SIX SUBSYSTEMS. ONE SAFE JOURNEY.

SAFETY LOOP CLOSURE · END-TO-END FLOW

01 Routing Engine

Prioritises safest paths over shortest distance by penalising high-risk segments derived from crime data.

02 Risk Scoring

Crimes targeting women receive maximum analytical weight, guaranteed circumvention of historically hostile streets.

03 Temporal Intelligence

Night-time multipliers redirect predictive paths heavily toward well-lit, trafficked thoroughfares.

04 SOS Emergency

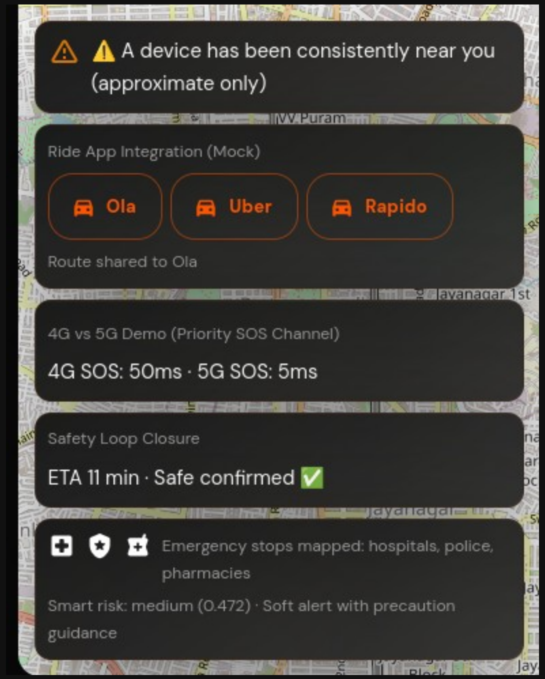
Long-press bypasses navigation to initiate live geo-tracking and transmit high-priority alerts to responders.

05 5G Safety Layer

Removes data-transmission limits, split-second distress beacons without buffer wait times.

06 Scalable Scope

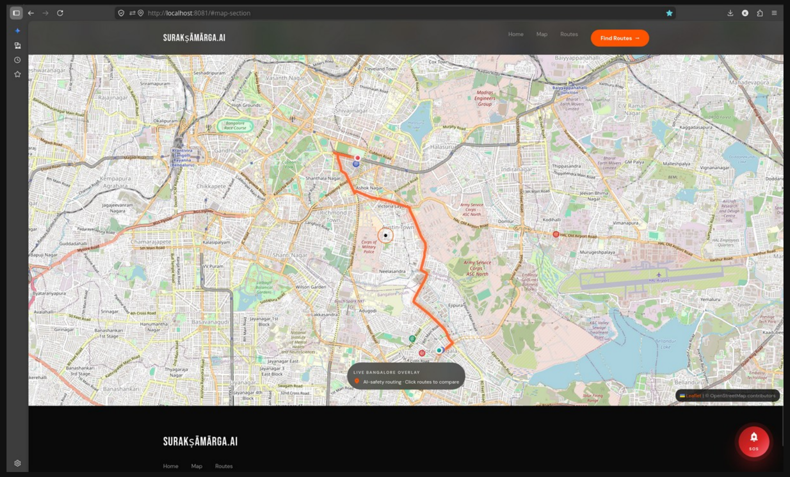
Demonstrated in Bengaluru on 157,160 records; architecture designed to expand to additional cities.



06 EXPERIMENTS & RESULTS

5G EDGE CHANGES EVERYTHING.

COMPLETE ROUTE MATRIX · PROTECTIVE INTERVENTIONS



EMERGENCY RESPONSE LATENCY

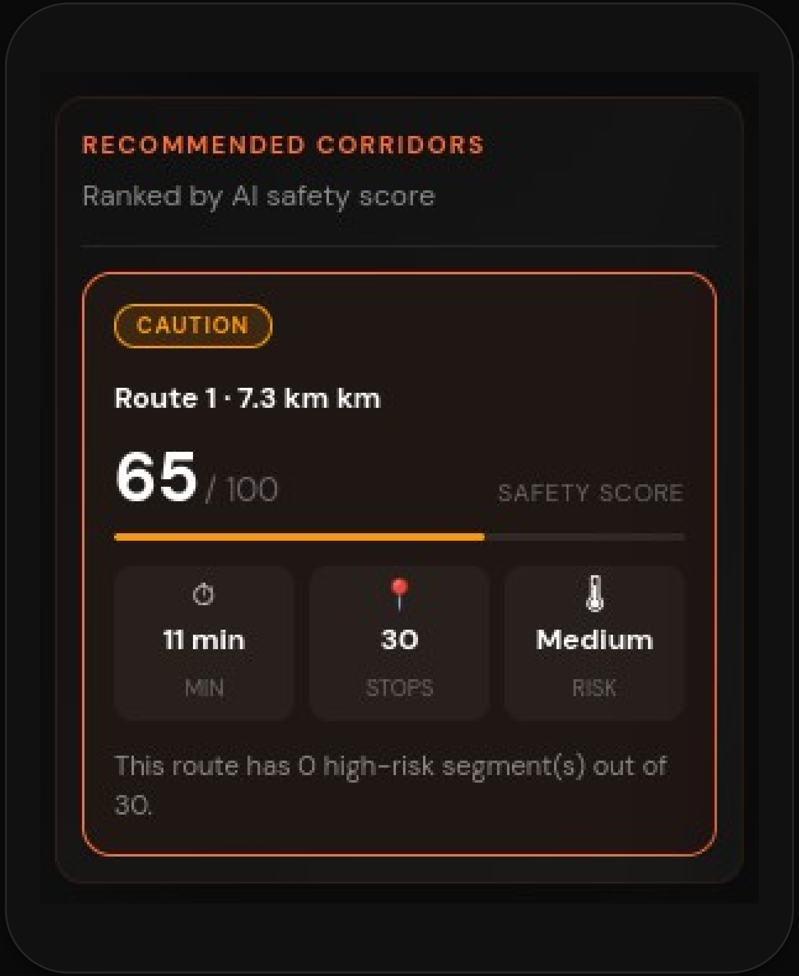
Scenario	3G	4G	5G Edge
Route Safety Scoring	~700 ms	~250 ms	**~50 ms**
SOS Distress Broadcast	~400 ms	~150 ms	**~10 ms**
Live Tracker Updates	Every 30 s	Every 15 s	**Every 2 s**

5G bypasses 4G connectivity ceilings, directly accelerating emergency response for women in transit.

07 NOVELTY

BUILT UNIQUELY FOR PROTECTION.

FIRST	**Women-first safety routing system.**
PRIORITY	**Safety prioritised over speed.**
INTEGRATED	**SOS + navigation in a single layer.**
REAL-TIME	**5G-enabled real-time protection.**



08 CONCLUSION

NAVIGATION, REBUILT AS SAFETY INFRASTRUCTURE.

By shifting the foundational routing algorithm to prioritise protection over proximity, SuraksaMarga.ai solves a critical real-world safety problem, transforming navigation from a convenience tool into real-time safety infrastructure for women.

FUTURE ENHANCEMENT

Real-time data integration enables predictive heat-mapping, anticipating risk zones dynamically by combining historical patterns with live situational awareness.